

On Problems Quantum Computers Can Solve Efficiently

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Outline

1. Quantum complexity theory
2. Representative algorithms (Grover's algo)
3. Sources of quantum advantage
4. Quantum machine learning bottlenecks.

Deterministic

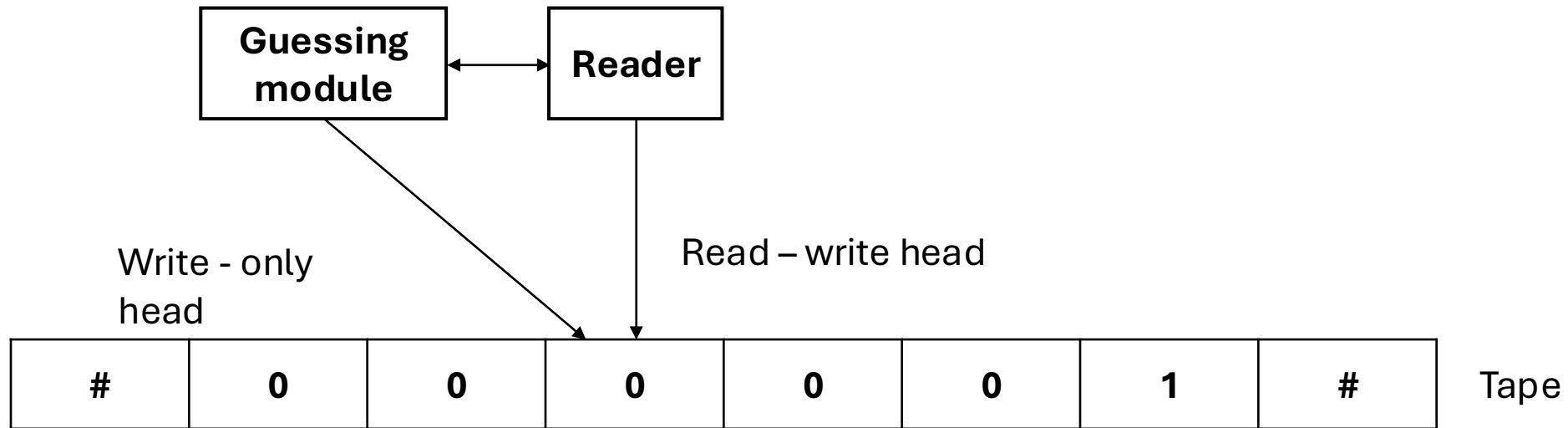
Deterministic Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$Q = \{States\}$ $\Sigma = \{Alphabet\} = \{0, 1, \#\}$

$\delta = Q \times \Sigma \rightarrow Q \times \Sigma \times \{L, R\}$ (transition function)

δ	0	1	#
q_{start}	q_L, L	q_{accept}	q_{reject}
q_L	q_L, L	q_{accept}	q_R, R
q_R	q_R, R	q_{accept}	q_{reject}

Nondeterministic

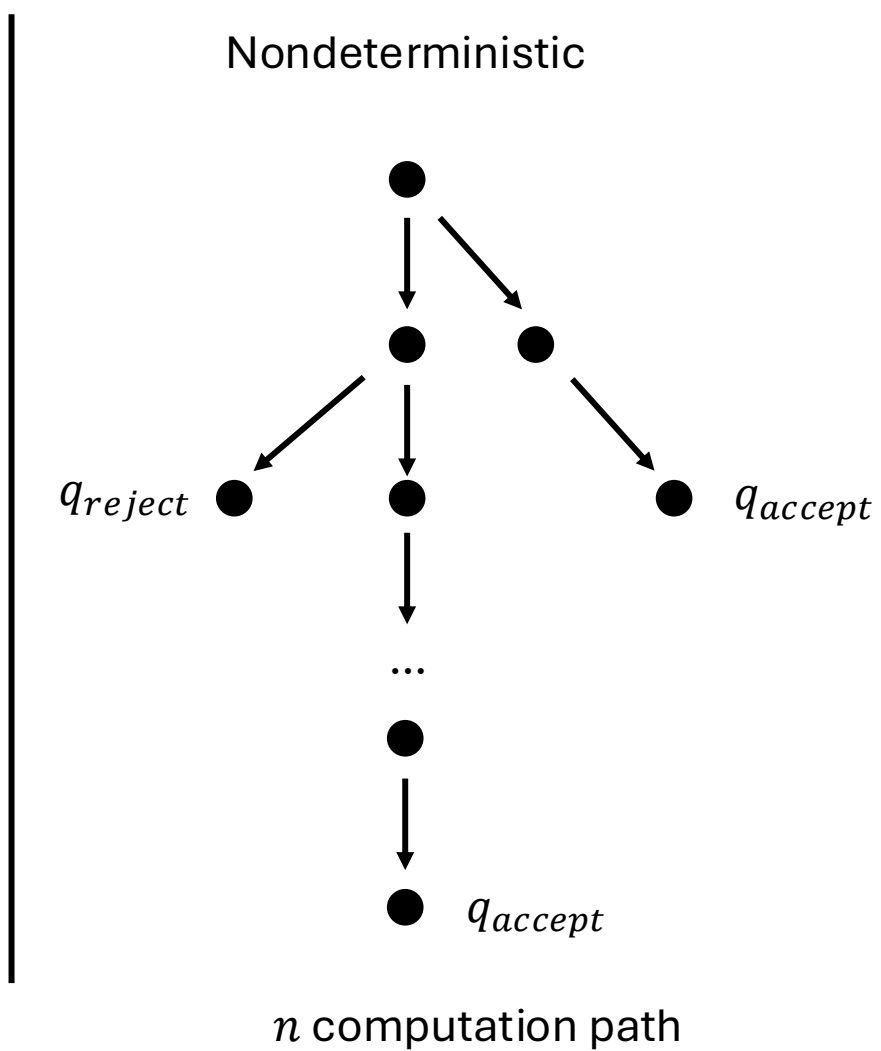
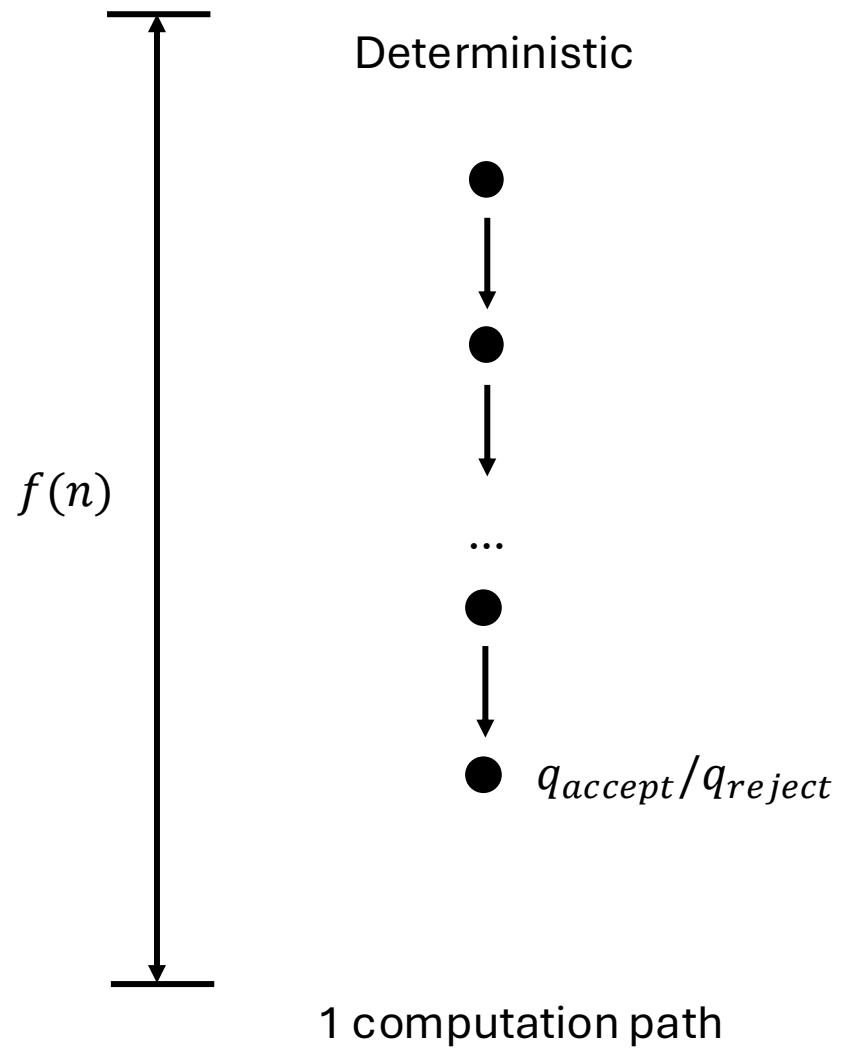


Nondeterministic

Nondeterministic Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$\delta = Q \times \Sigma \rightarrow \mathcal{P}(Q \times \Sigma \times \{L, R\})$, $\mathcal{P}(S)$ denotes for power set of S

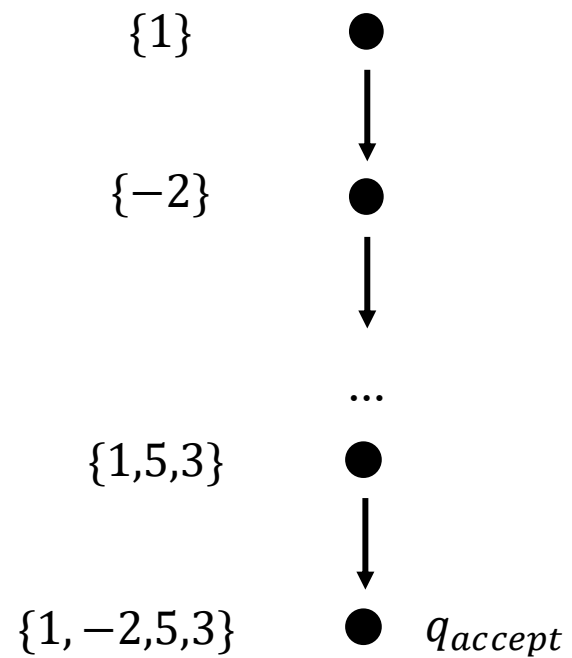
If the problem is solvable in polynomial time by a non-deterministic Turing machine, the problem belongs to NP class



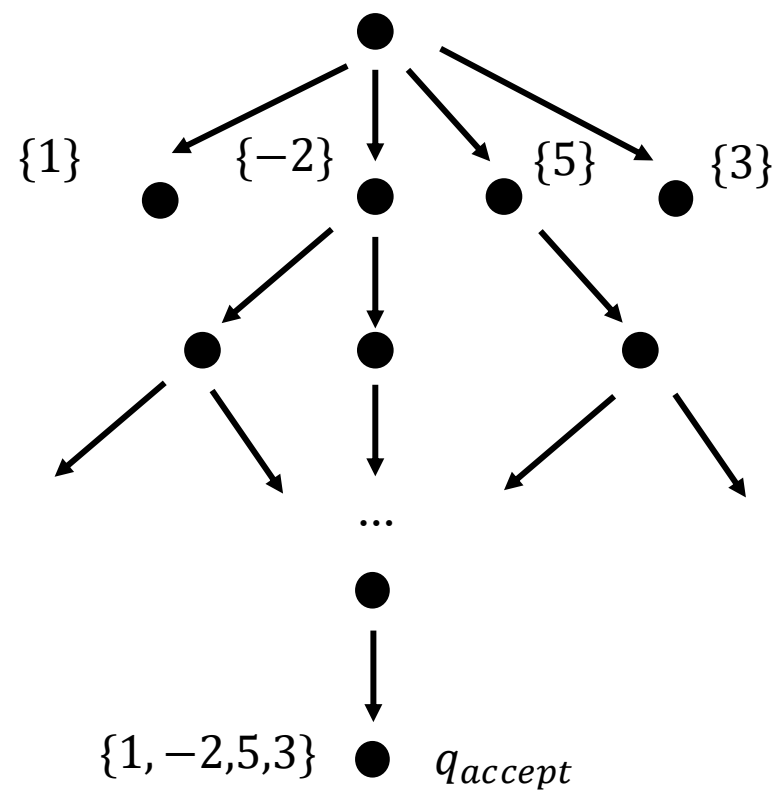
Example: Sub-set sum problem

Given a set $S = \{1, -2, 5, 3\}$. What sub-set of S has sum equal 7?

Deterministic



Nondeterministic



$S = \{1, -2, 5, 3\}$.

Polynomial number of **SPACE**

PSPACE

Kiểm tra lời giải trong thời gian đa thức

Nondeterministic Turing machine in Polynomial number of step

Cho một tập hợp các số nguyên, tồn tại hay không một tập hợp con có tổng bằng T không?

NP

coNP

complement of **NP**

Cho một tập hợp các số nguyên, có phải tất cả tập hợp con đều có tổng khác T không?

P = coP

Polynomial number of step

Tồn tại lời giải trong thời gian đa thức

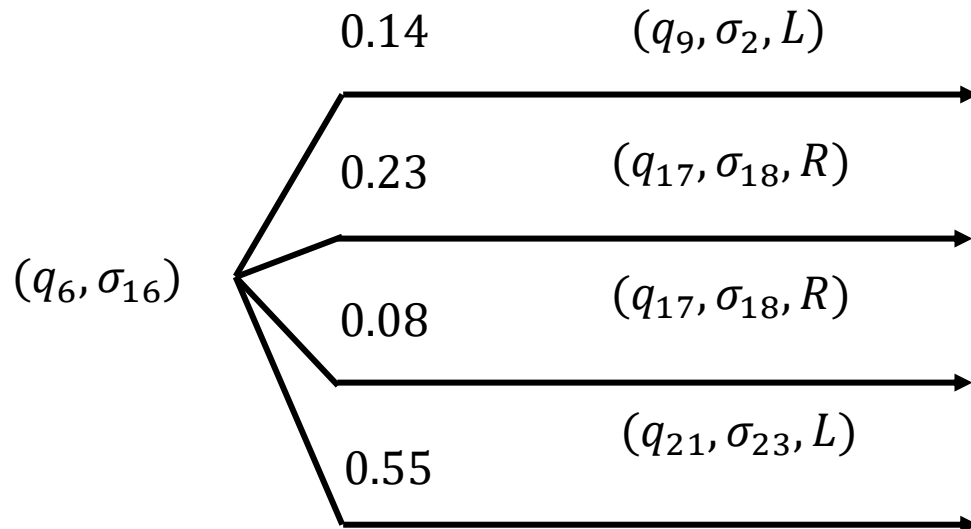
Probability

Nondeterministic Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$$\delta = Q \times \Sigma \rightarrow [0,1]^{(Q \times \Sigma \times \{L,R\})}$$

δ	0	1	#
q_{start}	$\frac{1}{2}: q_L, L;$ $\frac{1}{2}: q_R, R$	$1: q_{accept}$	$1: q_{reject}$
q_L	$1: q_L, L$	$1: q_{accept}$	$1: q_{reject}$
q_R	$1: q_R, R$	$1: q_{accept}$	$1: q_{reject}$

Probability



Lớp phức tạp. BPP là tập các vấn đề có thể được giải quyết bằng *PTM* (Probabilistic Turing Machine) trong thời gian đa thức (**P**olynomial time) với khả năng xảy ra lỗi. Để chính xác, nếu M là *PTM*, $L \in \mathbf{BPP}$ và x là input, thì:

$$x \in L \Rightarrow \text{Prob}(M \text{ accepts } x) > \frac{2}{3} \quad \text{Tính hội tụ!}$$

và

$$x \notin L \Rightarrow \text{Prob}(M \text{ rejects } x) > \frac{2}{3}$$

Lời giải đúng
Chấp nhận lời giải đúng

Lớp phức tạp. RP là tập các vấn đề có thể được giải quyết bằng *PTM* (Probabilistic / **R**andom Turing Machine) trong thời gian đa thức (**P**olynomial time) mà chỉ có khả năng sai tiêu cực. Nói cách khác, nếu M là *PTM*, $L \in \mathbf{RP}$ và x là input thì:

$$x \in L \Rightarrow \text{Prob}(M \text{ accepts } x) > \frac{2}{3}$$

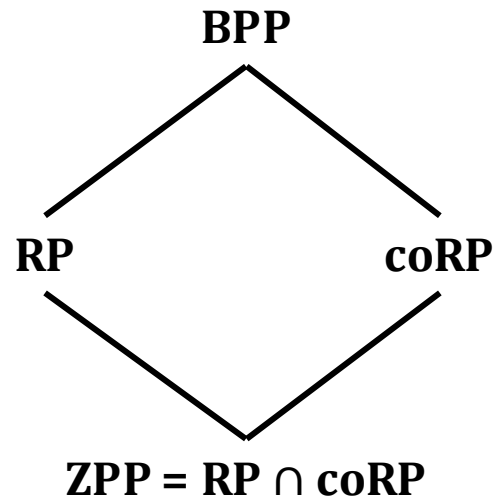
và

$$x \notin L \Rightarrow \text{Prob}(M \text{ rejects } x) = 1$$

Probability

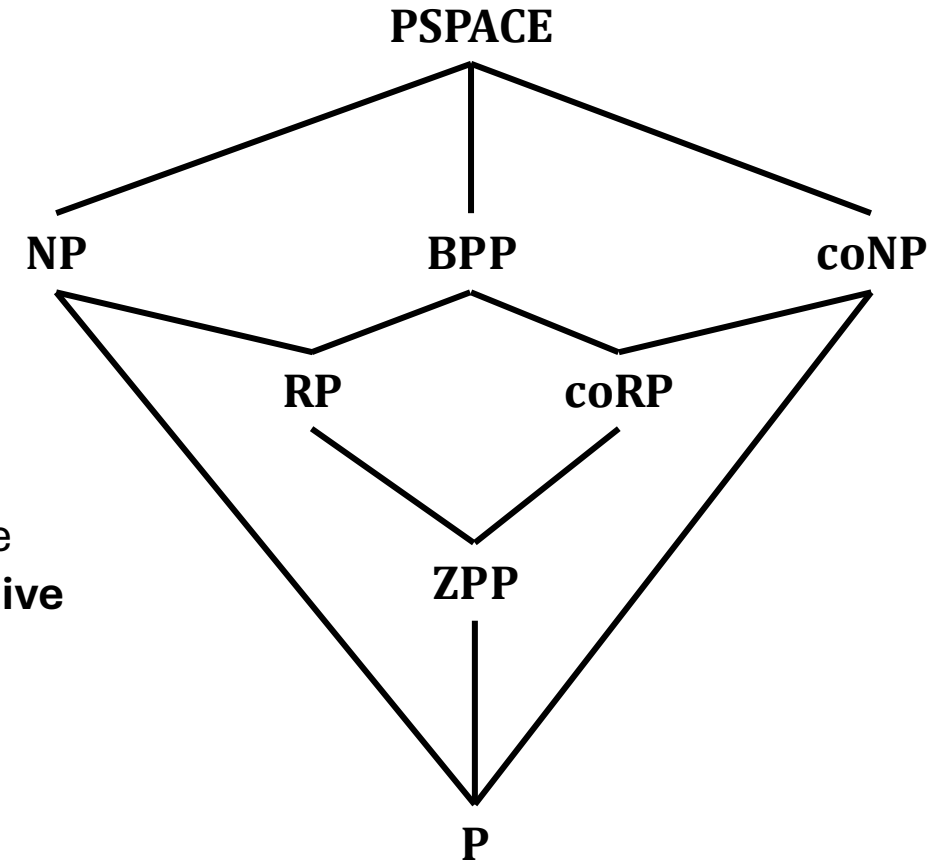
Probabilistic Turing
Machine in **P**olynomial
time với khả năng xảy ra lỗi

Probabilistic / **R**andom
Turing Machine in
Polynomial time with
false negative only



Probabilistic / Random Turing
Machine in **P**olynomial time
with **Z**ero error, allow ? result

Probabilistic /
Random Turing
Machine in
Polynomial time
with **false positive**
only



Quantum

Quantum Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$QTM = \{Q, \Sigma, q_{start}, q_{accept}, q_{reject}, \delta'\}$

$\delta': Q \times \Sigma \times Q \times \Sigma \times \{L, R\} \rightarrow \mathbb{C}$

$$(\forall q \in Q, \forall \sigma \in \Sigma) \sum_{q' \in Q, \sigma' \in \Sigma, D \in \{L, R\}} |\delta'(q, \sigma, q', \sigma', D)|^2 = 1$$

Quantum

δ	0	1	#
q_{start}	$\frac{1}{\sqrt{2}}: q_L, L;$ $\frac{1}{\sqrt{2}}: q_R, R$	1: q_{accept}	1: q_{reject}
q_L	1: q_L, L	1: q_{accept}	1: q_{reject}
q_R	1: q_R, R	1: q_{accept}	1: q_{reject}

$$| \#000q_{start}0010\# \rangle \mapsto \frac{| \#00q_L00010\# \rangle + | \#0000q_R010\# \rangle}{\sqrt{2}}$$

$$\mapsto \frac{| \#0q_L000010\# \rangle + | \#00000q_R10\# \rangle}{\sqrt{2}}$$

$$\mapsto \frac{| \#q_L0000010\# \rangle + | \#00000q_{accept}10\# \rangle}{\sqrt{2}}$$

Quantum

$$EQP \subseteq ZQP \subseteq BQP$$

Simon, Grover

Lớp phức tạp. BQP là tập các vấn đề có thể được giải quyết bằng *QTM* (Quantum Turing Machine) trong thời gian đa thức (**P**olynomial time) với sai số (**B**ounded error) bị chặn cả hai biên. Nói cách khác, nếu M là *QTM*, $L \in BQP$ và x là input, thì

$$x \in L \Rightarrow \text{Prob}(M \text{ accepts } x) > \frac{2}{3}$$

và

$$x \notin L \Rightarrow \text{Prob}(M \text{ rejects } x) > \frac{2}{3}$$

Same as ZPP

Lớp phức tạp. ZQP là tập các vấn đề có thể được giải quyết bằng *QTM* (Quantum Turing Machine) trong thời gian đa thức (**P**olynomial time) mà không xảy ra lỗi (**Z**ero error). Nói cách khác, nếu M là *QTM*, $L \in ZQP$ và x là input, thì có một cơ hội nhỏ hơn 50% *QTM* dừng ở trạng thái "không biết" (do not know), trường hợp ngược lại:

$$x \in L \Rightarrow \text{Prob}(M \text{ accepts } x) = 1$$

và

$$x \notin L \Rightarrow \text{Prob}(M \text{ rejects } x) = 1$$

Deutsch-Jozsa,
Require only 1
query

Lớp phức tạp. EQP là tập các vấn đề có thể được giải quyết bằng *QTM* (Quantum Turing Machine) trong thời gian đa thức chính xác (**P**olynomial time **E**xactly) (không xảy ra lỗi). Nói cách khác, nếu M là *QTM*, $L \in EQP$ và x là input, thì:

$$x \in L \Rightarrow \text{Prob}(M \text{ accepts } x) = 1$$

Và

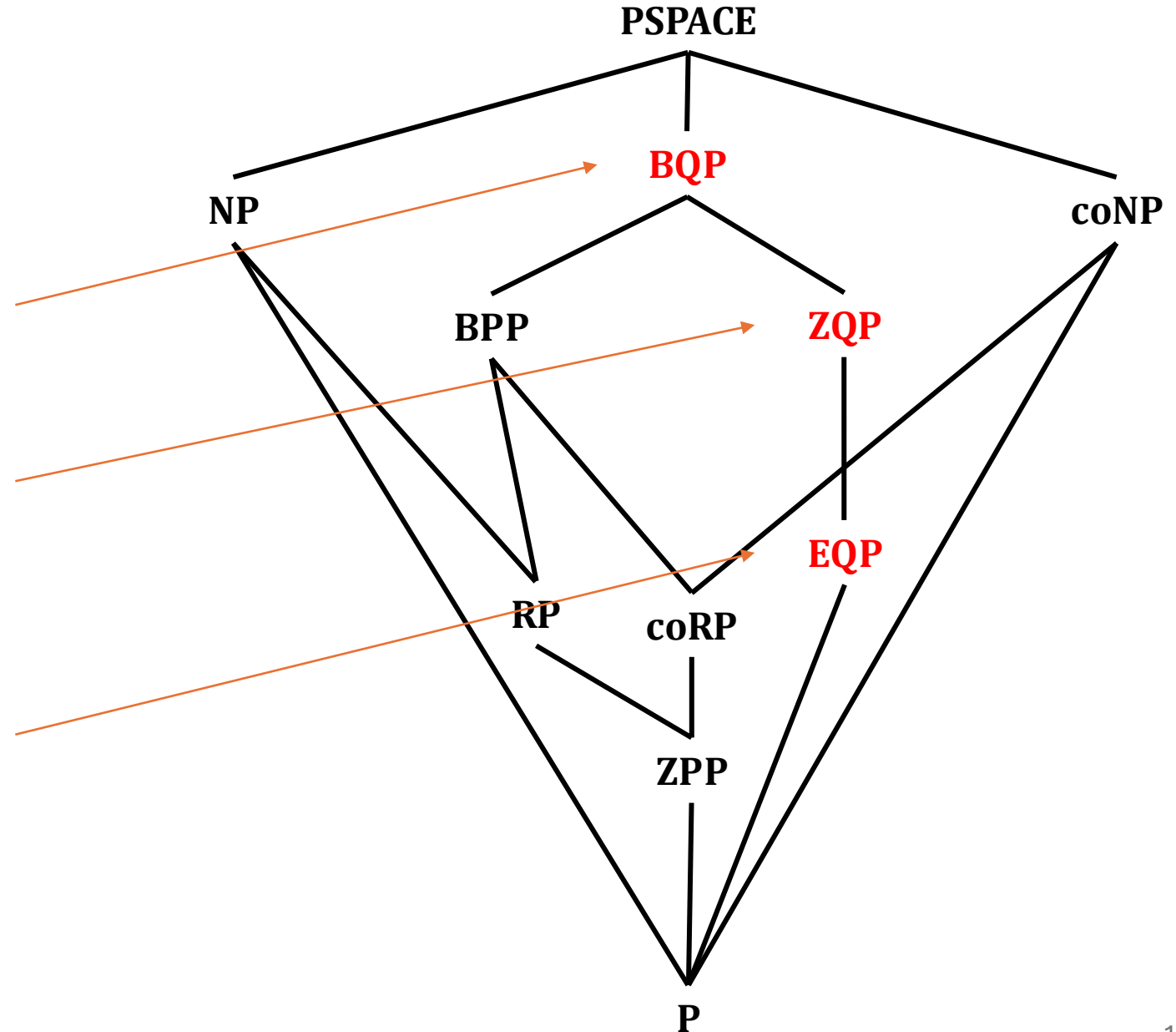
$$x \notin L \Rightarrow \text{Prob}(M \text{ rejects } x) = 1$$

Quantum

Quantum Turing Machine in
Polynomial time with **B**ounded error
(sai số bị chặn)

Quantum Turing Machine in
Polynomial time with **Z**ero error

Quantum Turing Machine in
Polynomial **E**xactly time with no
error



2. Grover algo

Spatial search (discrete time)

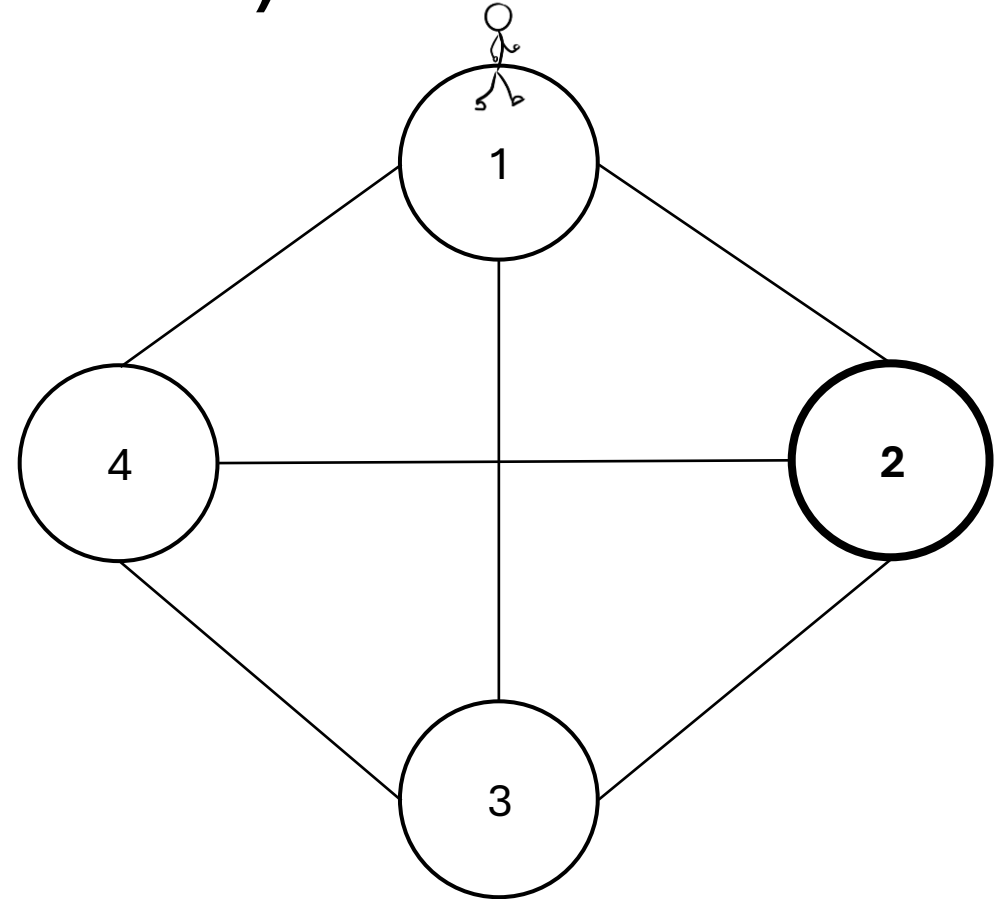
Assume that we have a complete graph with 4 nodes.

One approach is using random walk.

Note that:

Walker must be in somewhere

$$\Rightarrow \sum P_j = 1$$

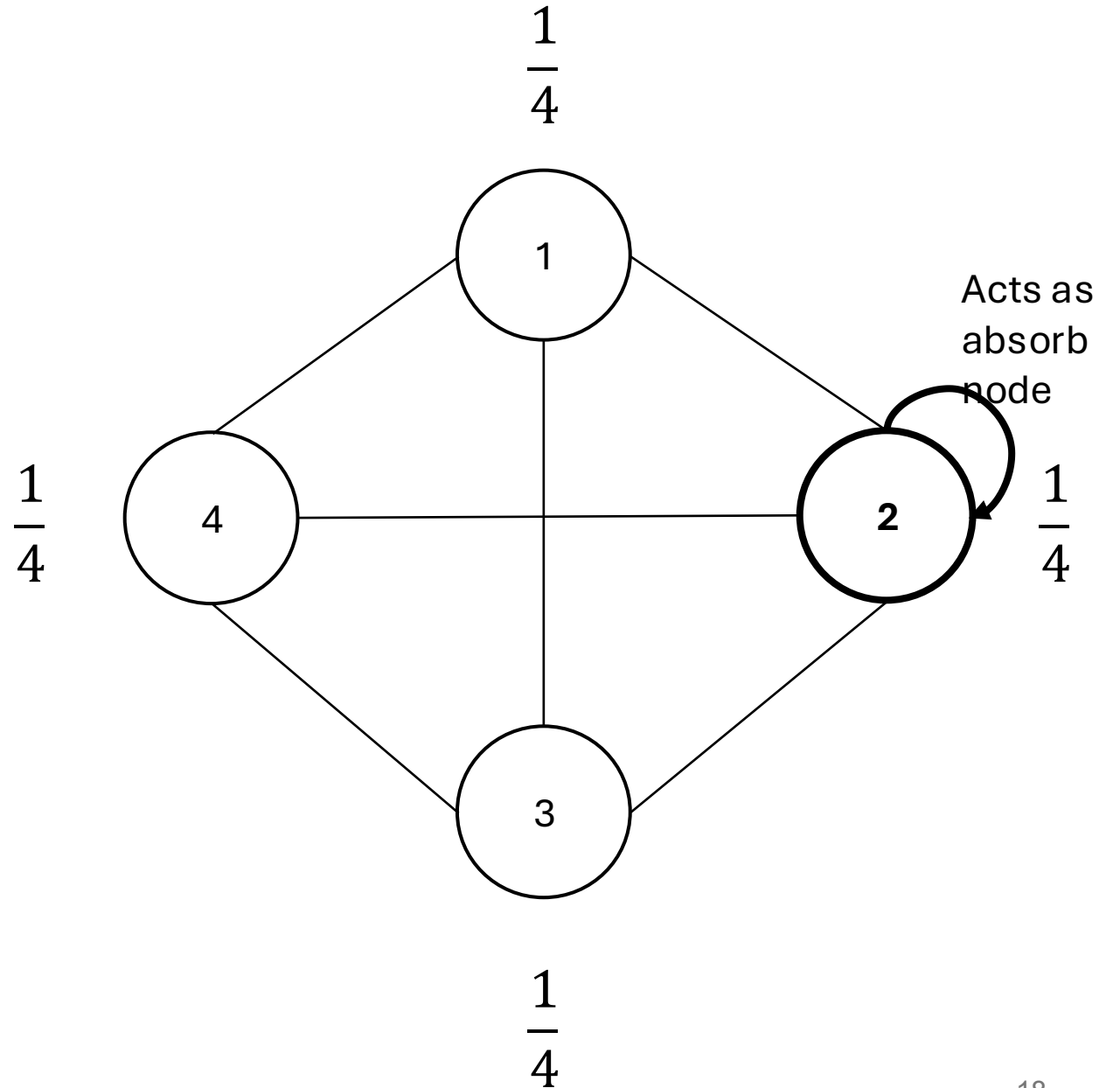


Random walk

Initial state: $\frac{1}{4}$ at all node

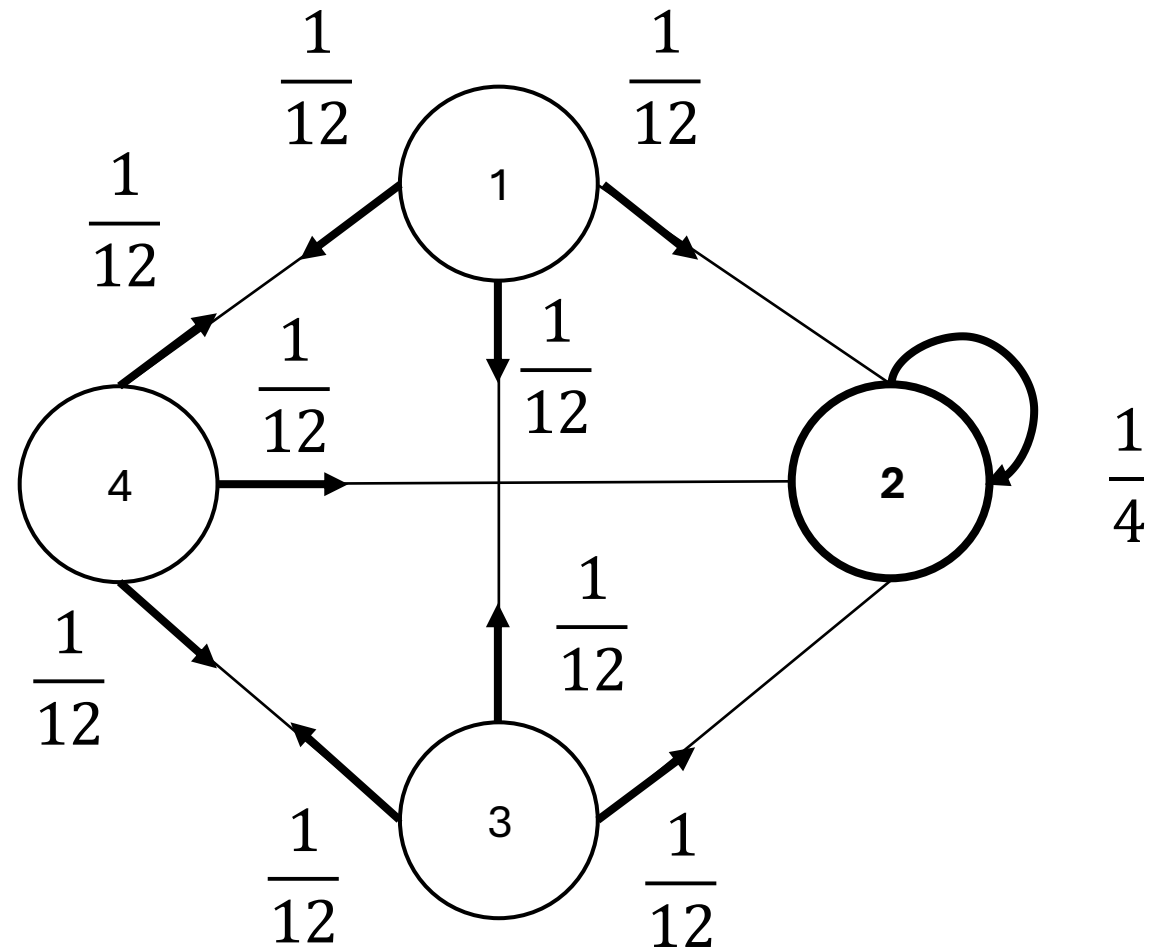
At step k : jump to other random nodes

$$P_1 = P_2 = P_3 = P_4 = 1/4$$



Random walk

At step 1:

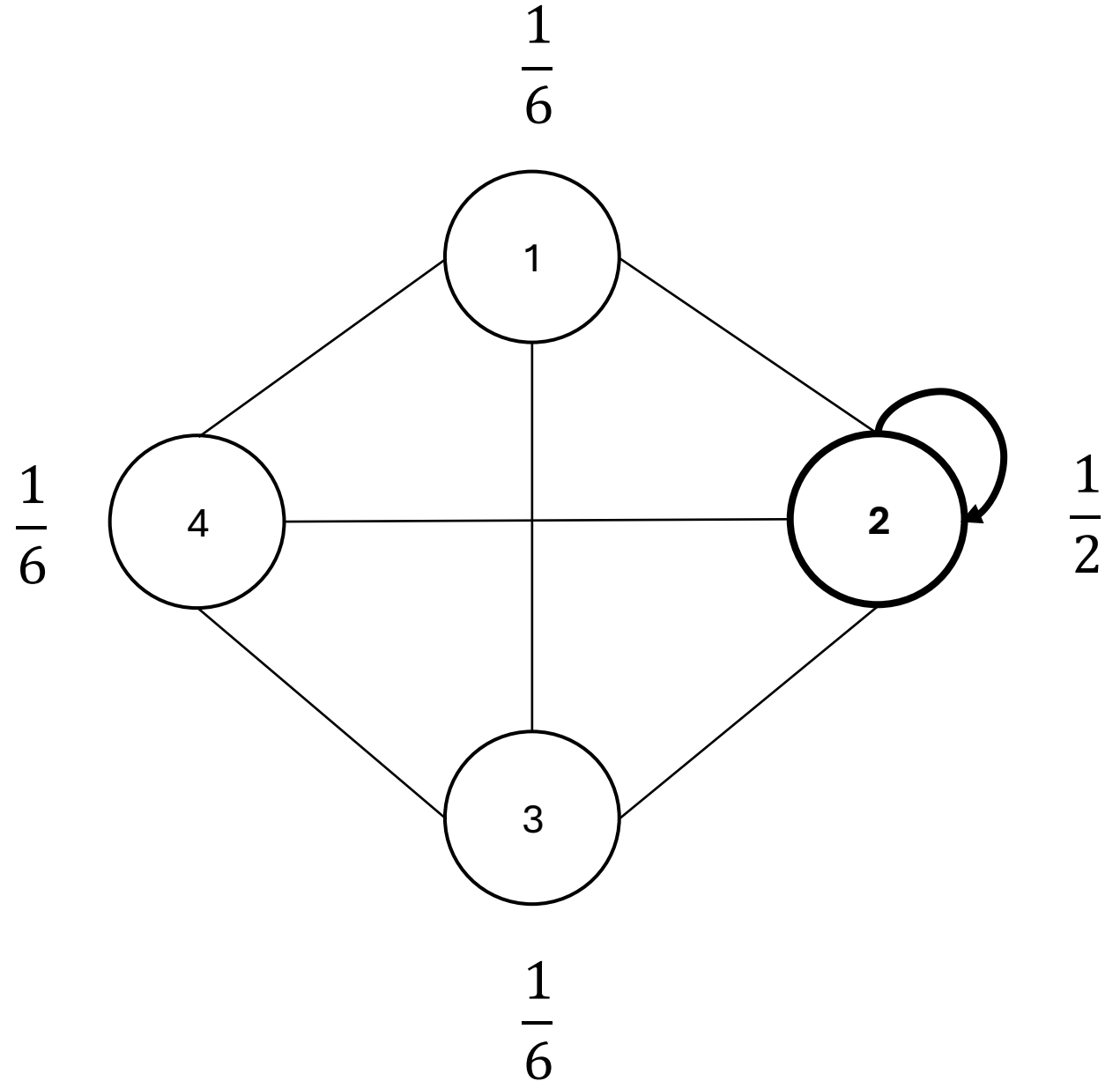


Random walk

When done step 1:

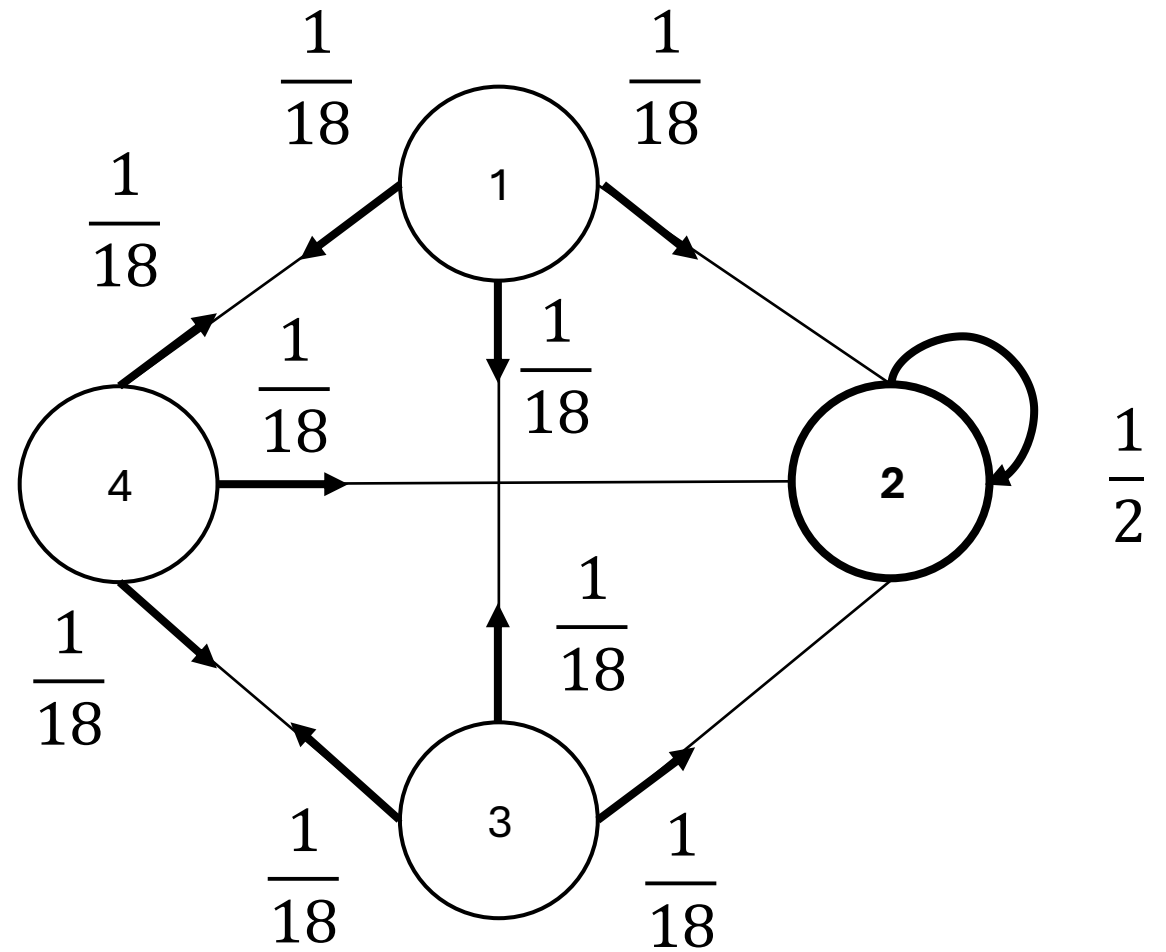
$$P_1 = P_3 = P_4 = \frac{1}{12} * 2 = \frac{1}{6}$$

$$P_2 = \frac{1}{4} + \frac{1}{12} * 3 = 1/2$$



Random walk

At step 2:

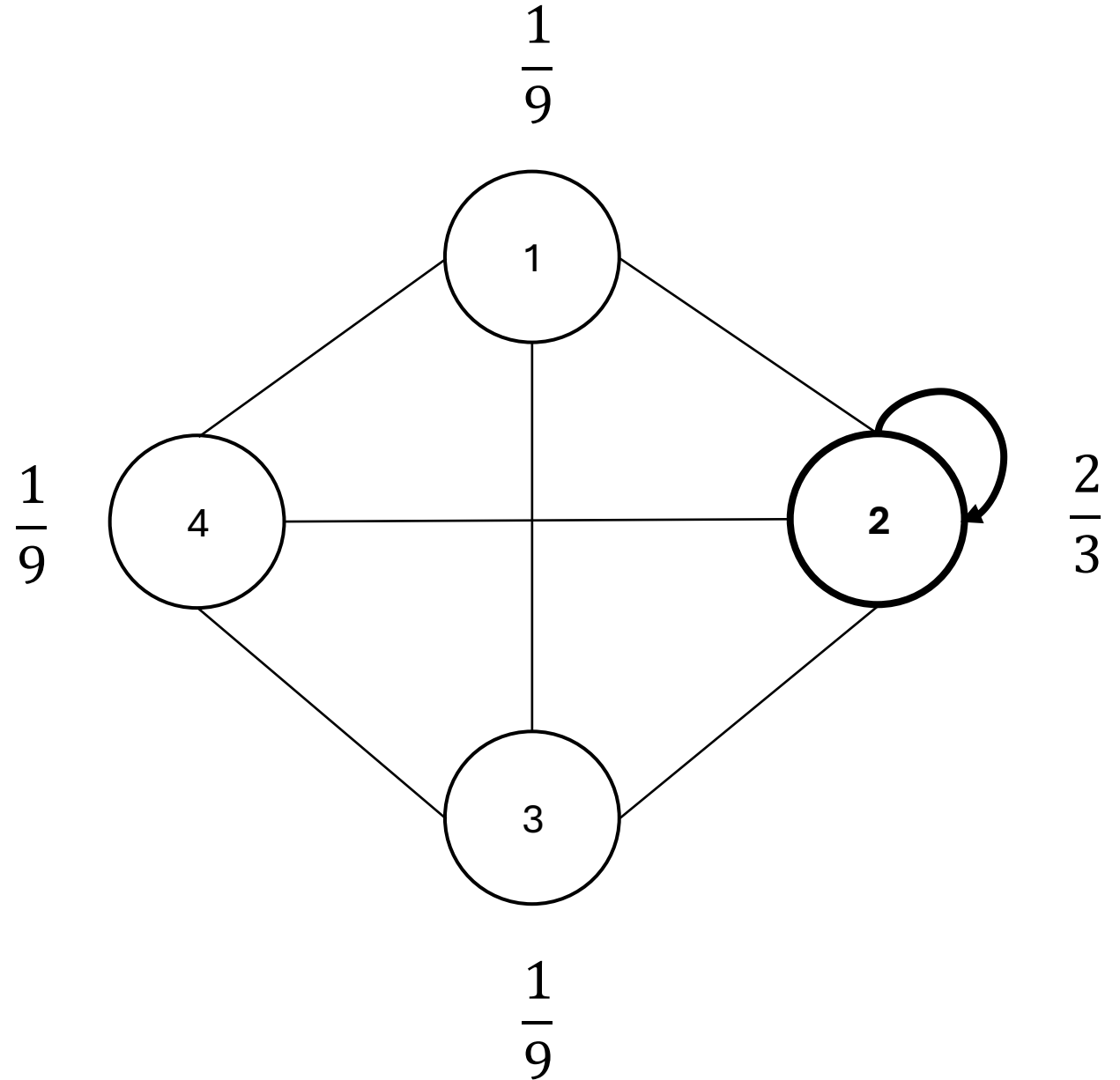


Random walk

When done step 2:

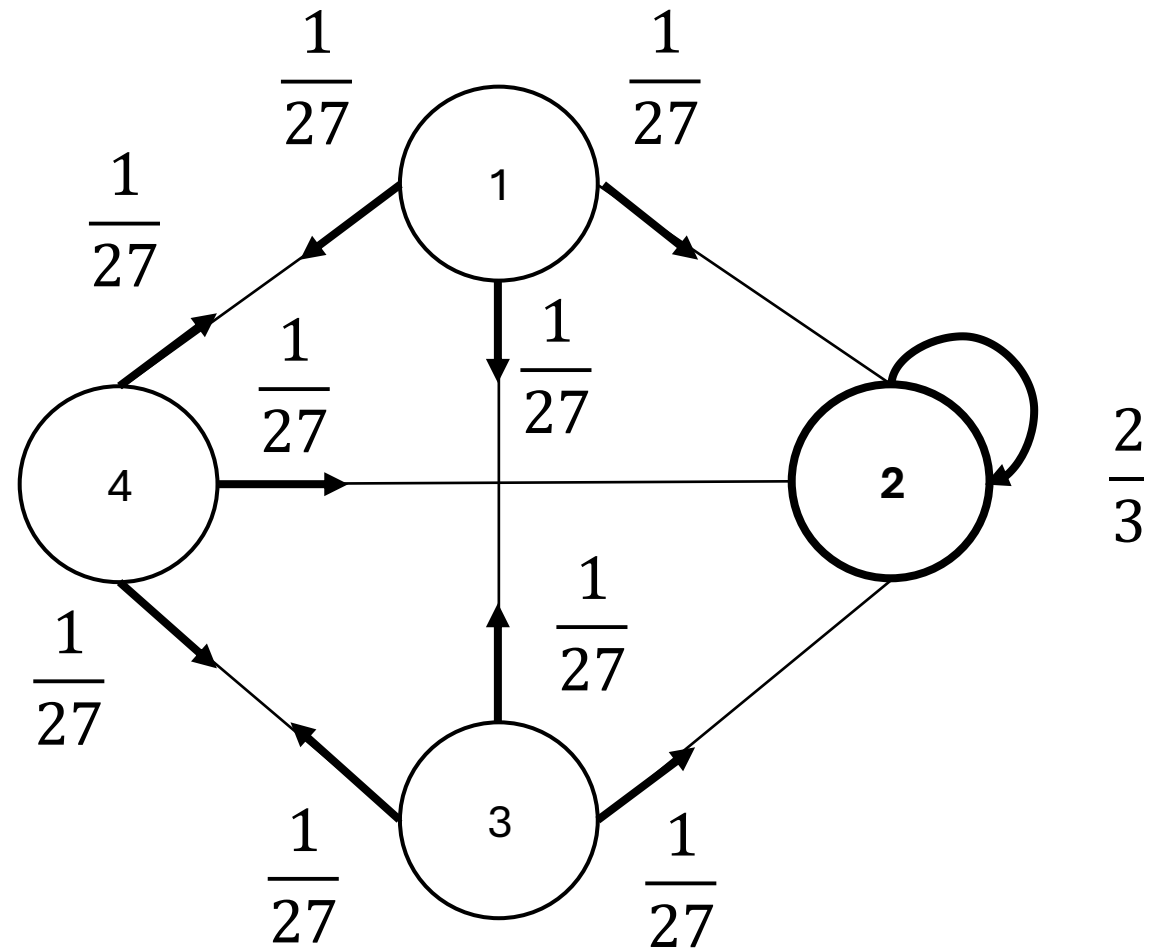
$$P_1 = P_3 = P_4 = \frac{1}{18} * 2 = \frac{1}{9}$$

$$P_2 = \frac{1}{2} + \frac{1}{18} * 3 = \frac{2}{3}$$



Random walk

At step 3:

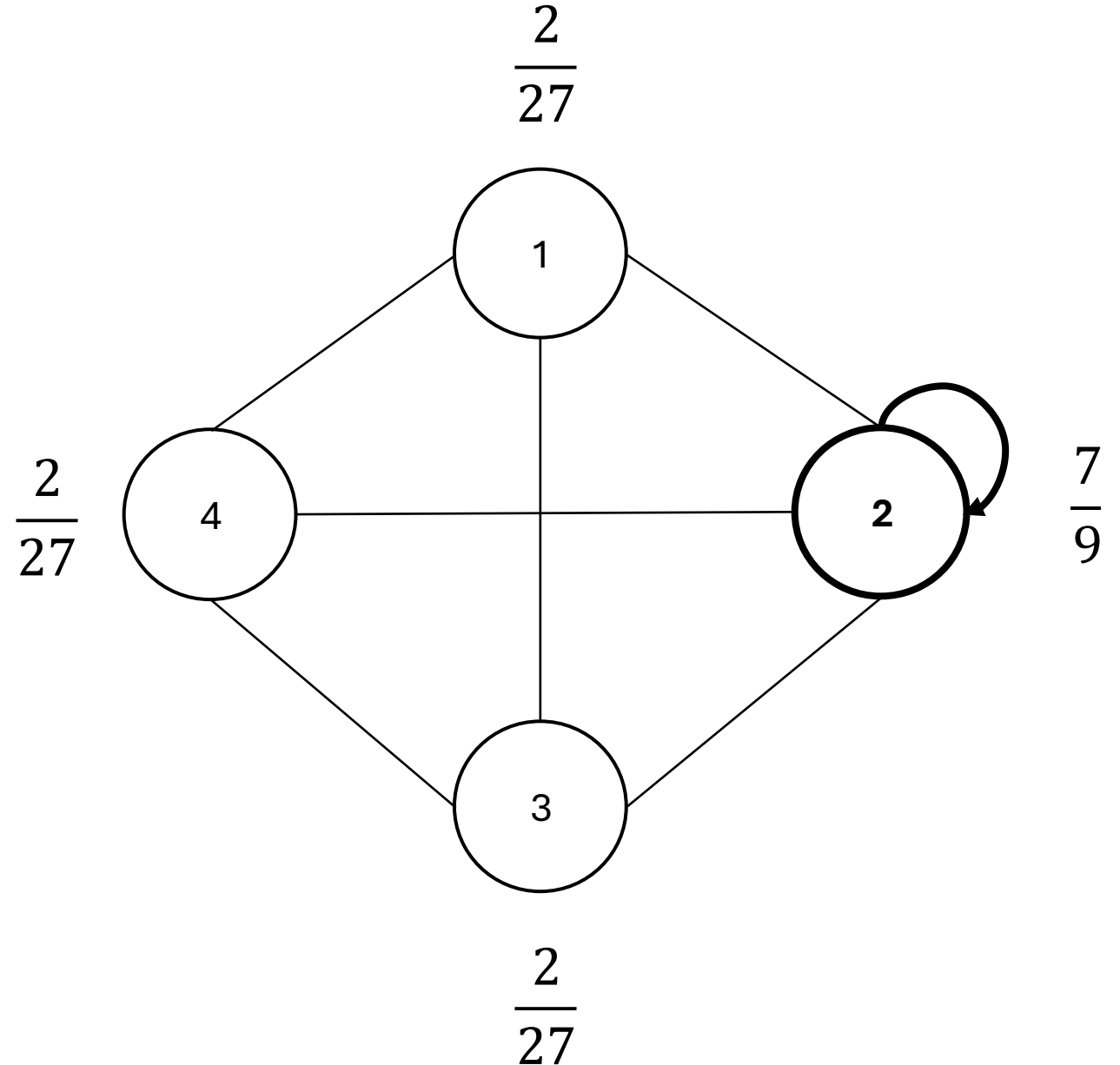


Random walk

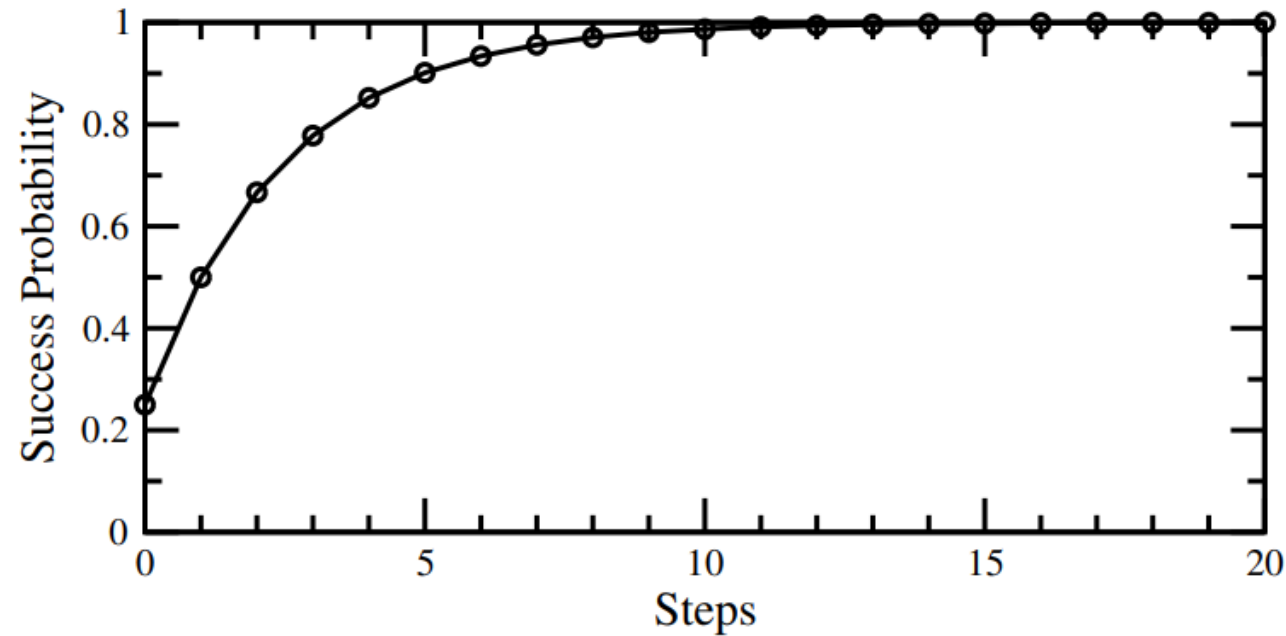
When done step 3:

$$P_1 = P_3 = P_4 = \frac{1}{27} * 2 = \frac{2}{27}$$

$$P(2) = \frac{2}{3} + \frac{1}{27} * 3 = \frac{7}{9}$$



Random walk



Prob(Success) at time $t =$

$$1 - \frac{N-1}{N} \left(\frac{N-2}{N-1} \right)^t$$

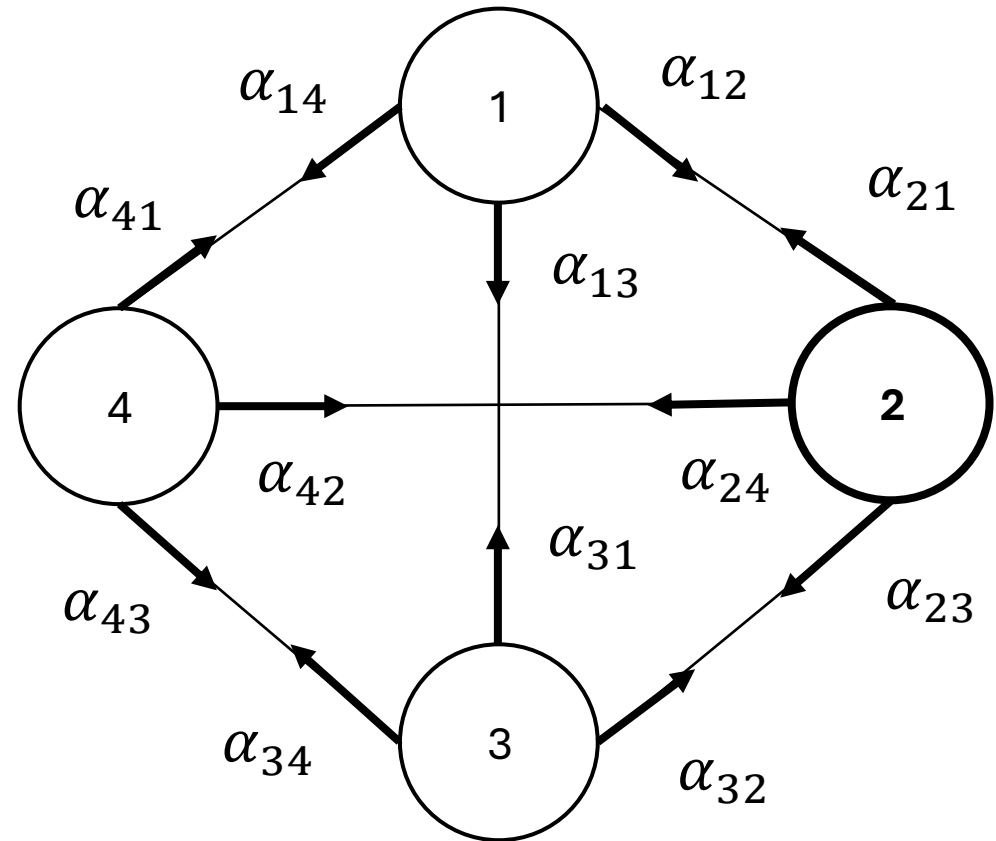
Runtime: $O(N)$

Quantum walk

Change 4 probs to 12 amplitudes of a wave function $|\psi\rangle := \{\alpha_i\}$

Init: $|\psi(0)\rangle$

Note that at intermediate steps we cannot observe the system. Then we do not know the current node is the marked node or not.



At node 2: $A = \frac{-1}{\sqrt{12}} * \frac{3}{3}$

Quantum walk

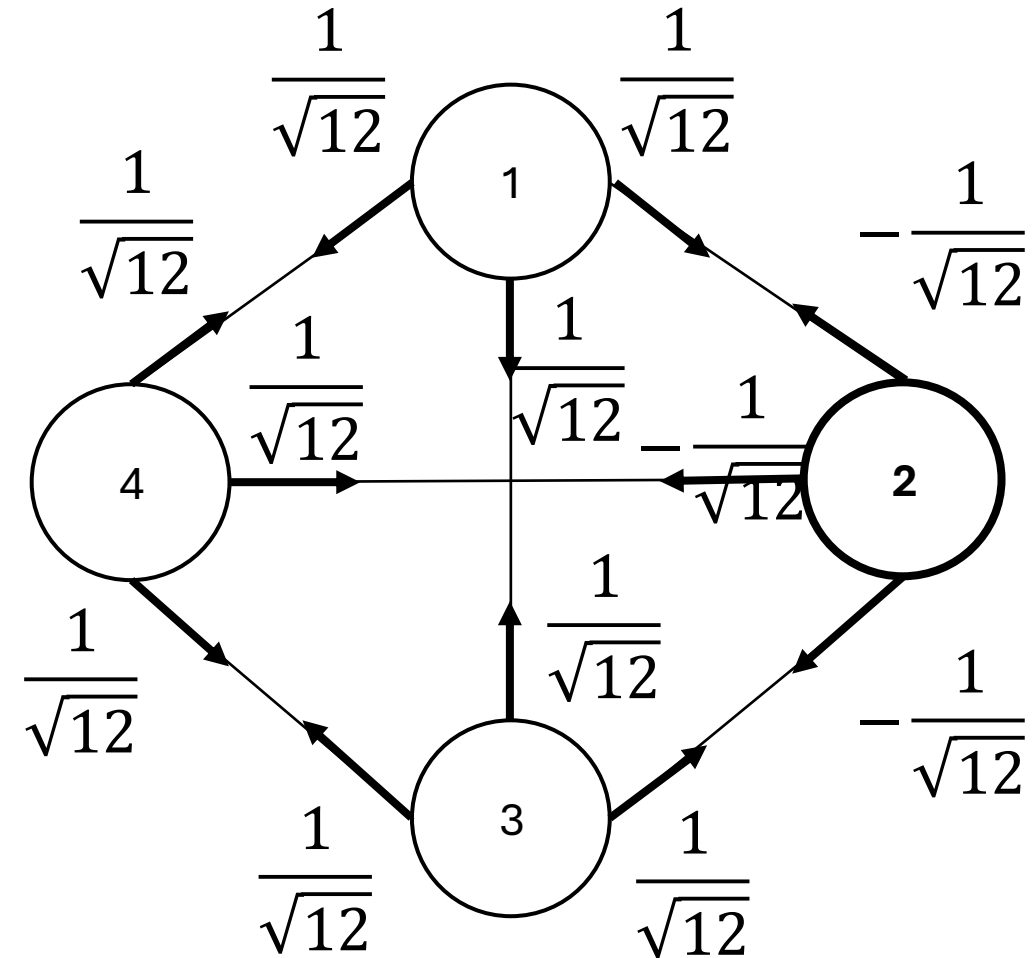
At step 1

1a. Query oracle:

$$Q|\psi(0)\rangle$$

1b. Apply a quantum coin flip+shift $(-I+2A)$ that inverts each amplitude about the average amplitude at its vertex. (Nothing happen)

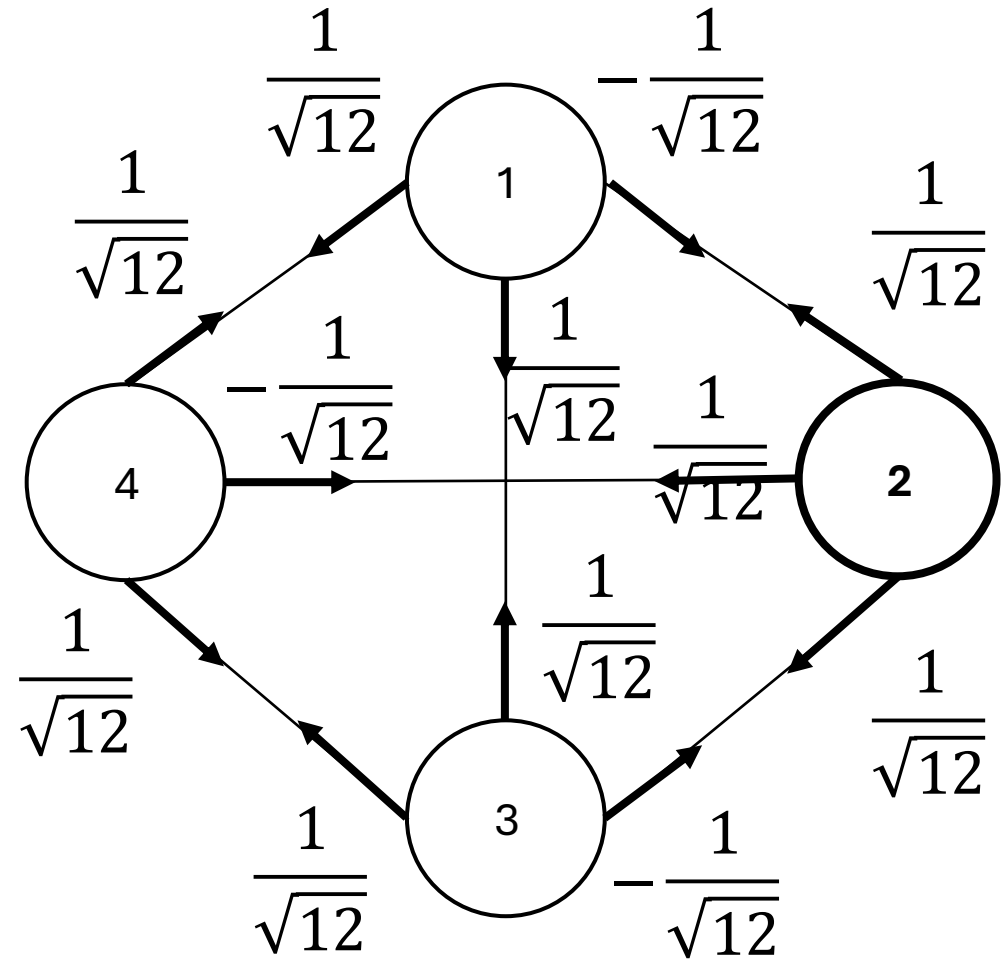
$$C(Q|\psi(0)\rangle)$$



Quantum walk

1c. Shift operator (walk):
walker at vertex i pointing to
vertex j to hop to vertex j and
point back to vertex $i \Rightarrow$
Swap α_{ij} with α_{ji} .

$$SCQ|\psi(0)\rangle = |\psi(1)\rangle$$

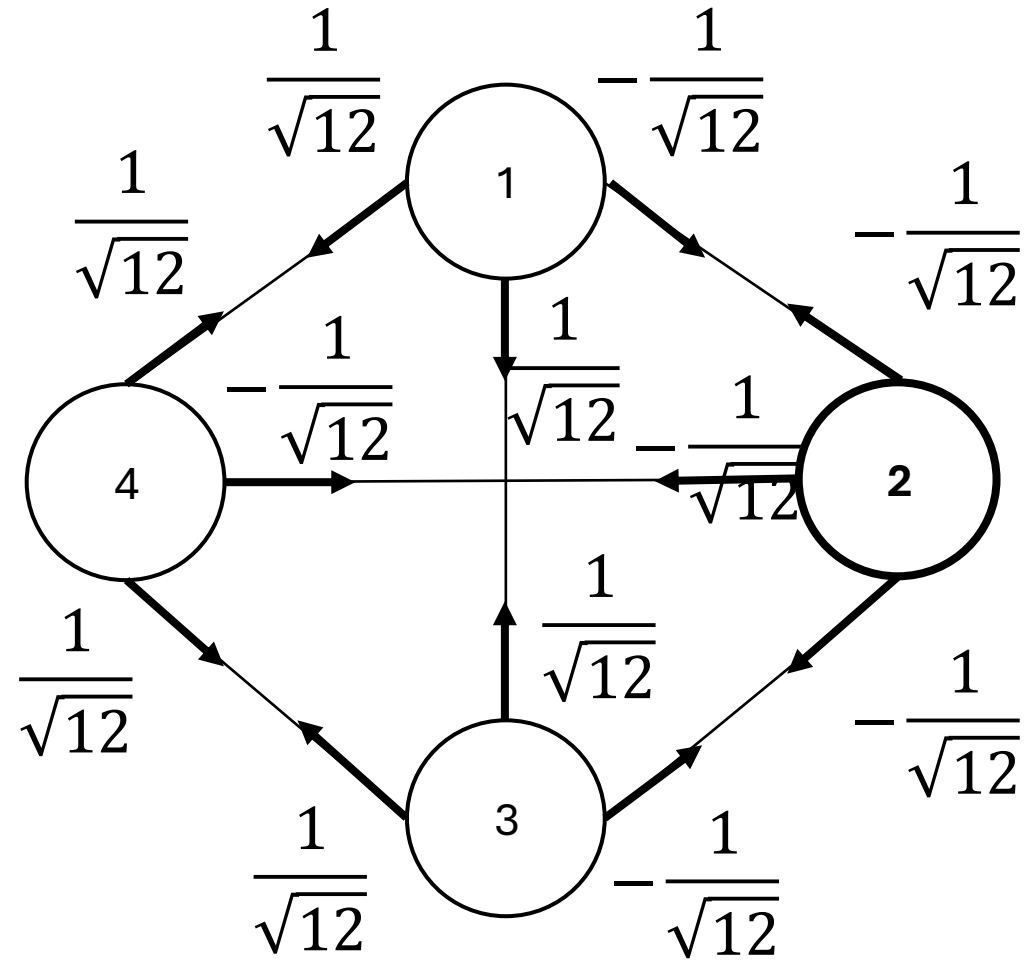


Quantum walk

At step 2

2a. Query oracle:

$$Q|\psi(1)\rangle$$



Quantum walk

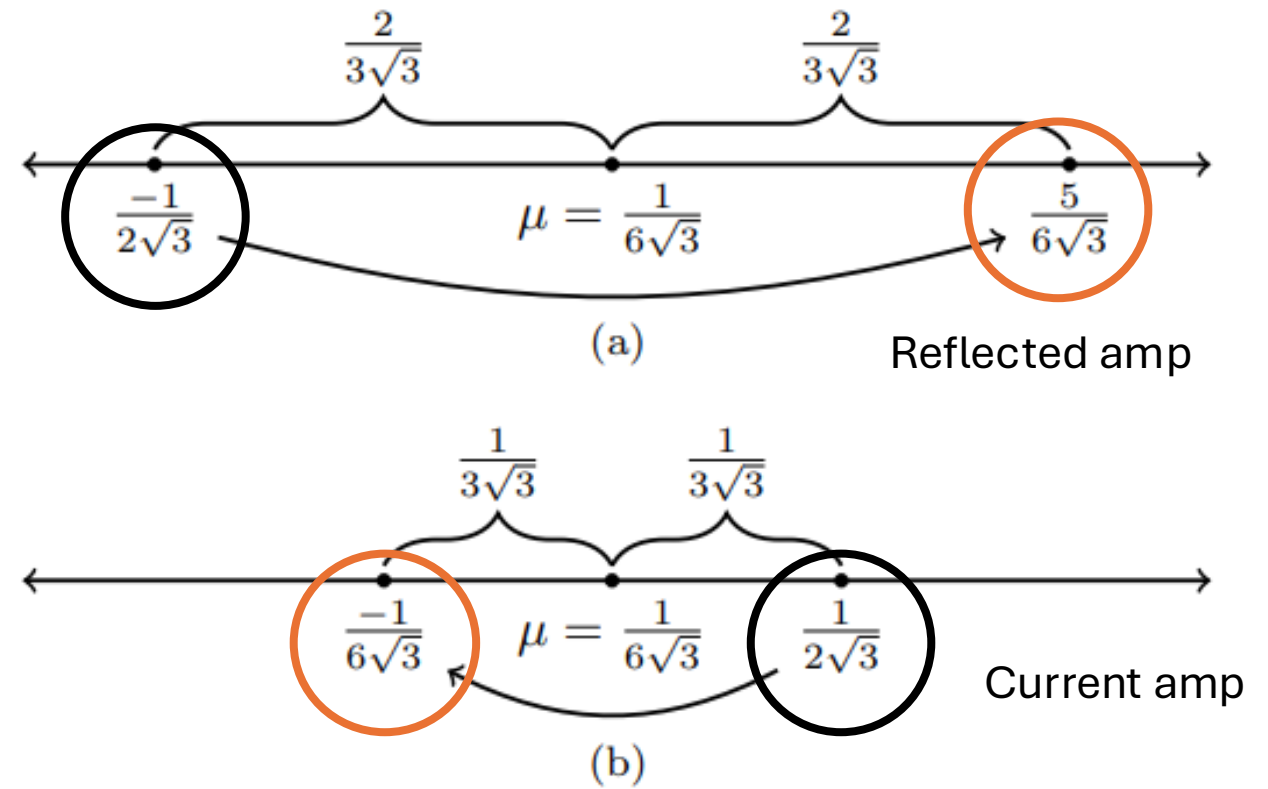
2b. The coin inverts or reflects each amplitude about their average.

$(-I+2A)$

$$\text{At node 1: } A = \frac{\frac{1}{\sqrt{12}} + \frac{1}{\sqrt{12}} - \frac{1}{\sqrt{12}}}{3} = \frac{1}{6\sqrt{3}}$$

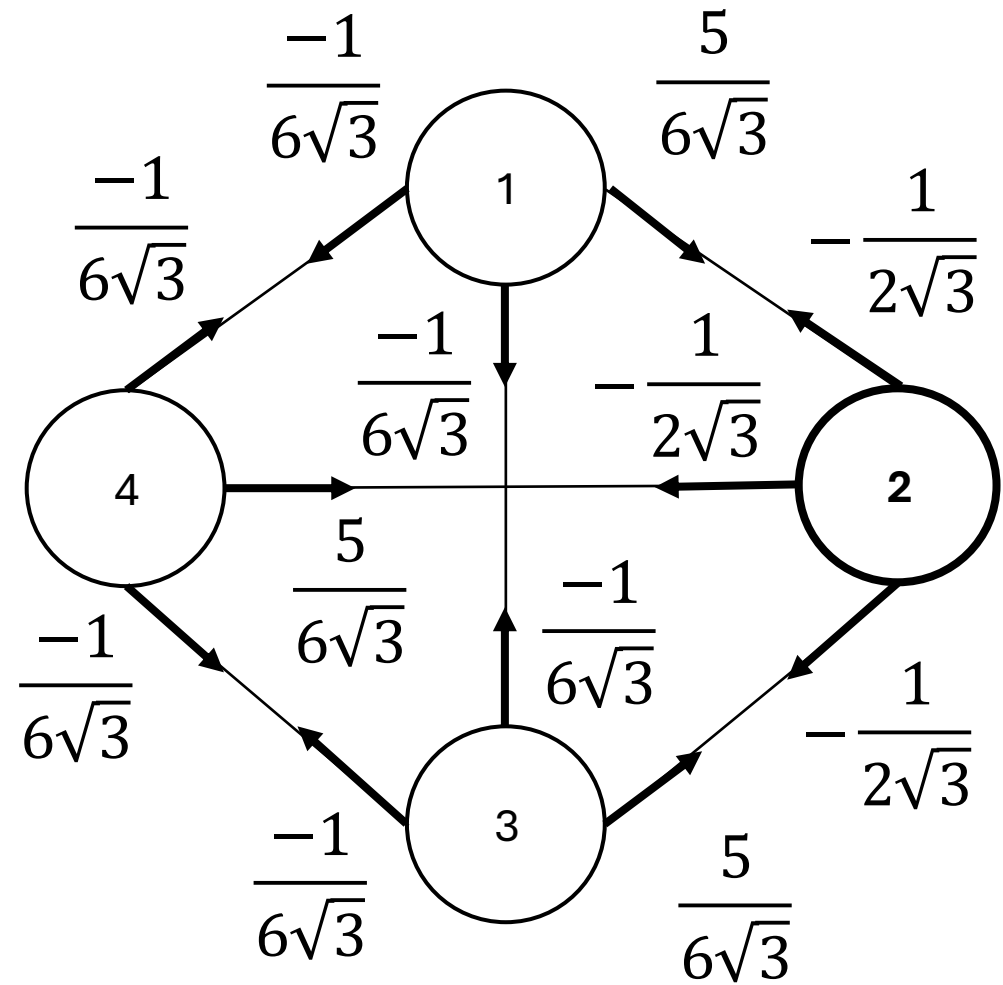
$$\Rightarrow \alpha_{12} = -\alpha_{12} + 2A = \frac{5}{6\sqrt{3}}$$

$$\Rightarrow \alpha_{13} = -\alpha_{13} + 2A = \frac{-1}{6\sqrt{3}}$$



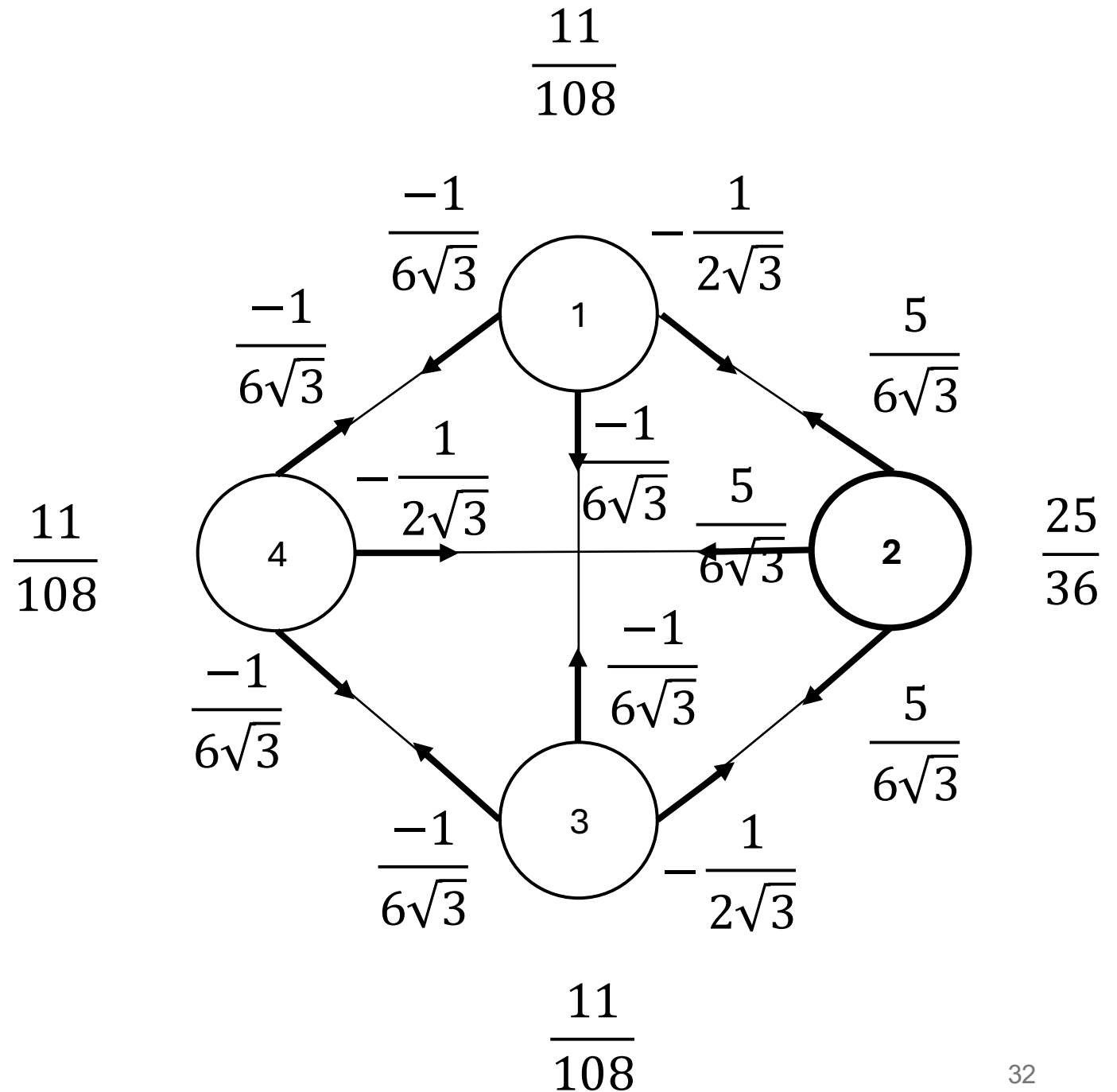
Quantum walk

2c. Shift operator (walk):
swap α_{ij} with α_{ji} .

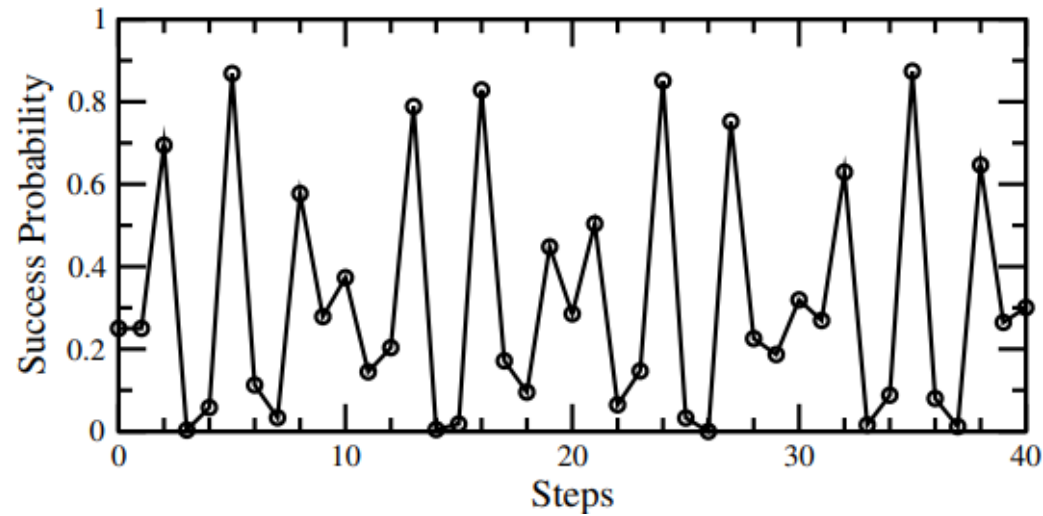


Quantum walk

2c. Shift operator (walk):
swap α_{ij} with α_{ji} .



Quantum walk



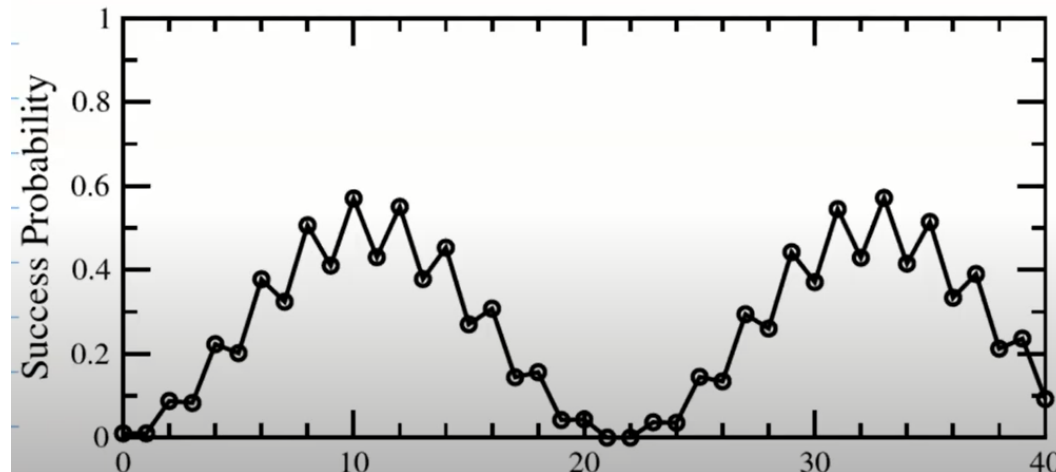
N=4

Prob(Success) at time $t =$

$$\left\{ (N-1) \left[\cos(\phi t) + \sqrt{2N-3} \sin(\phi t) \right] + (-1)^t (N-2) \right\}^2 / [(2N-3)^2 N],$$

where $\phi = \sin^{-1} \left(\frac{\sqrt{2N-3}}{N-1} \right)$

Runtime: $O(\sqrt{N})$



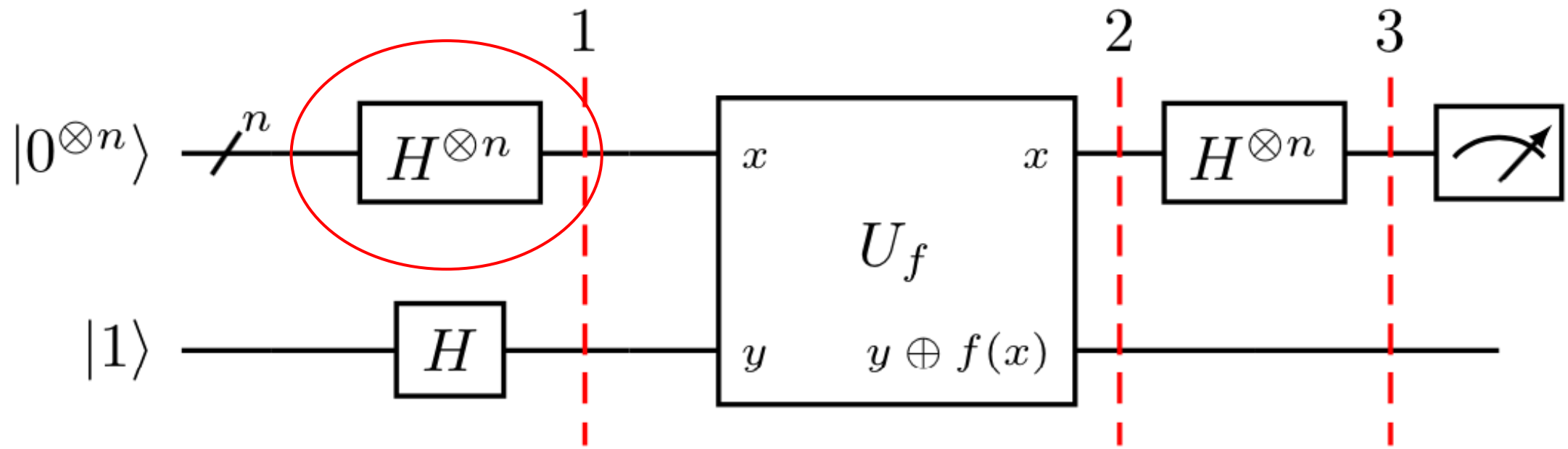
N=10

0

3. Sources of quantum advantage

Largely it comes from superposition.

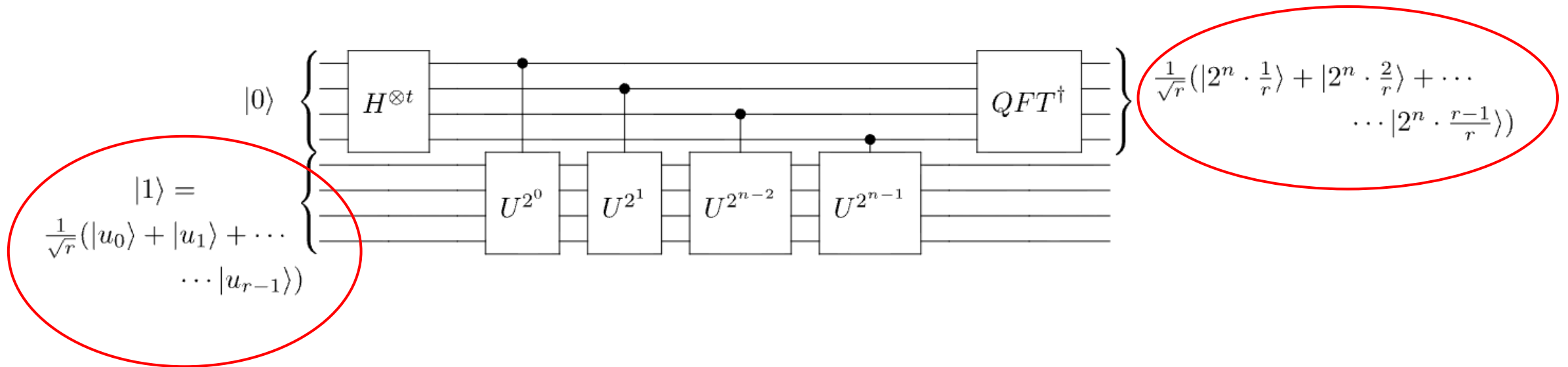
Look Deutsch-Josza algorithm



3. Sources of quantum advantage

Largely it comes from superposition.

Look Shor's algorithm



3. Sources of quantum advantage

Total runtime = $T_{prepare}$ (build a useful representation) + T_{solve} (find the solution).

Recent quantum algorithms: $T_{prepare} \gg T_{solve}$

The right benchmark is the strongest available classical workflow, heuristics included - not an idealized worst case.

Q: “Can quantum give speedups under the same heuristic assumptions classical methods use?”

3. Source of Advantage: example in quantum simulation

Classical

- Must store d^n complex amplitudes explicitly.
- Matrix–vector multiplication scales as $O(d^{2n})$.

• Quantum

- Represents the same Hilbert space with only n qubits.
- Under efficient block encodings, cost can scale as $O(\text{poly}(n))$; storage becomes $O(n)$.

3. Source of Advantage: locality and light cone

We usually solving the eigenvalue problem:

$$\hat{H}c = Ec$$

1. $\hat{H}$ is sparsity => ok, solvable on classical computer
2. But $\hat{H}^k$ generate a light-cone structure. We must to use resources for saving correlation information
 $L_c \sim nR_c$, where n is the number of iterations and R_c denotes the interaction range. Consequently, $N_{\text{corr}} \sim (nR_c)^{mD} = L_c^{mD}$ where D is the physical dimension of the system.

The resulting quantum advantage is approximately $\frac{T_C}{T_Q} \sim \frac{L_c^{mD}}{T_{BE}}$,

Bottleneck: Data Preparation

1. Classical data must first be converted into quantum state
2. A N – dim classical vector may need only $\mathcal{O}(\log N)$ qubits.
3. Without QRAM, state-preparation cost can scale badly enough to eliminate any theoretical speedup entirely.
4. Block encoding: QSVT and Hamiltonian simulation
assume a unitary that embeds a matrix - constructing it is often non-trivial and costly for dense matrices.

4. Limitations of Quantum Machine Learning

1. QML is one of the most actively studied topics, with many claimed advantages over classical ML.
2. Small-scale evidence: reported gains often come from small experiments or idealized simulations, not current noisy hardware.
3. Hidden data cost: many claimed speedups neglect the cost of data preparation, building the block encoding can cost more than solving the problem classically.
4. Idealized access: complexities like $\mathcal{O}(\text{poly log } n)$ frequently assume access models that hide real data-acquisition and encoding costs.

Which problems do quantum computers solve efficiently?

- 1. A quantum speedup must be judged end-to-end — not by the quantum portion alone.**
2. Quantum simulation, condensed matter physics, and quantum chemistry offer the most near-term opportunities.
3. For QML, the bottleneck is typically converting classical data into quantum-ready form, not the circuit itself.

What Will QCs Be Good For?

- Simulating quantum physics and chemistry
- Breaking current public-key encryption
- We'll need to get lucky for most other applications
(The world has often been lied to about this)

Looking for quantum advantage

What problems do people hope to solve with classical simulation? Let's use quantum!



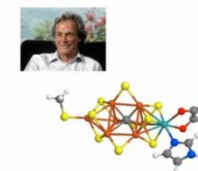
So What Are The Main Known Applications of QCs?

1. Breaking Current Public-Key Cryptography



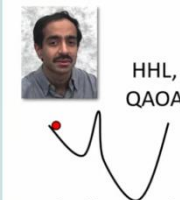
Requires fault-tolerance
Post-quantum crypto is a viable response

2. Simulating Quantum Physics and Chemistry



Still the best known "killer app." Some near-term hope?

3. Everything Else...



HHL, QAOA
Looks like mostly modest speedups and **not** near-term, but who knows?